

Practice Quiz: Action / Renormalization Group

(Advanced Theory)

Name: __ Date: __

Part A: Multiple Choice

1. The renormalization group relates: A) Descriptions at the same scale B) Descriptions at different scales — systematically integrating out fine-grained degrees of freedom to produce effective coarse-grained descriptions C) Only quantum systems D) Only classical systems
2. Active Inference claims to be scale-free because: A) It ignores scale B) The free energy functional has the same mathematical form at every RG scale — from molecules to societies — with scale-specific parameters C) It only applies to brains D) Scale is irrelevant
3. Neural criticality is supported by: A) Only fMRI studies B) Power-law avalanche distributions, $1/f$ noise, long-range correlations, and maximal dynamic range and information capacity C) Proving the brain is a crystal D) EEG alone
4. Nested Markov blankets mean: A) Blankets are identical B) Lower-scale blankets (cell membranes) are contained within higher-scale blankets (organ boundaries), creating a hierarchy of inferential agents C) Only one blanket exists D) Blankets are flat
5. RG coarse-graining corresponds to: A) Random deletion B) Bayesian Model Reduction — removing unnecessary parameters while preserving essential structure of the generative model C) Adding complexity D) Data collection
6. RG fixed points correspond to: A) Static systems B) Scale-invariant descriptions exhibiting critical phenomena — power laws, long-range correlations, and maximal sensitivity C) Frozen dynamics D) Equilibrium only
7. Universality classes in RG theory explain why: A) All systems are identical B) Vastly different physical systems (magnets, fluids, neural networks) share the same critical exponents

— their behavior near criticality depends only on symmetry and dimensionality, not microscopic details C) Only one system matters D) Criticality is rare

8. The relationship between Active Inference and the FEP at different scales is: A) Unrelated across scales B) The FEP provides the scale-invariant principle, while Active Inference specifies the scale-dependent implementation — like the relationship between a physical law and its manifestation in a particular material C) Only valid at the neural scale D) A metaphor without formal content

Part B: Short Answer

1. A neuroscience lab measures neuronal avalanche size distributions in cortical slices and finds a power law with exponent $\alpha \approx 1.5$. Explain what this finding suggests about the brain's operating point, how it connects to Active Inference, and name one confound that could produce a spurious power law. (200 words)

2. Consider a three-level hierarchy: cell $\rightarrow$ tissue $\rightarrow$ organism. For each transition, identify what degrees of freedom are "integrated out" in the RG sense and what effective parameters remain. How does the Markov Blanket change at each scale? (200 words)

Part C: Essay Questions

1. Perform a detailed RG analysis of a multi-scale biological system (your choice: immune system, ecosystem, neural circuit). Identify at least 3 scales, the states at each scale, the coarse-graining procedure between scales, and whether the effective Active Inference equations preserve their form. Where does the analysis work well? Where does it break down? (500 words)

2. Evaluate the "brain at criticality" hypothesis. Present the strongest evidence for and against. How would you test whether the brain is: (a) at a critical point, (b) near but not at a critical point, or (c) somewhere else entirely? What would each finding imply for Active Inference? (400 words)

3. Write a theoretical essay on how Active Inference applies to social systems. Choose a specific social phenomenon (political polarization, market dynamics, cultural evolution) and

model it using scale-free Active Inference. Identify the generative model, the prediction errors, the active inference policies, and the Markov blanket. What novel predictions does this framework generate? (500 words)